POVM

Suppose we pick a set of m_{max} operators Ω_m, $\sum_{m=1}^{m_{max}} \Omega_m^\dagger \Omega_m = \mathbb{1}$.

$$\Longrightarrow \quad \rho_f = \frac{\Omega_m \rho \Omega_m^\dagger}{Tr(\Omega_m \rho \Omega_m^\dagger)}, \quad P(m) = Tr(\Omega_m \rho \Omega_m^\dagger)$$

POVMs : positive operator-valued measurements.

$$P(m \in [a,b]) = \sum_{m=a}^{b} Tr(\Omega_m \rho \Omega_m^\dagger) = Tr\left(\sum_{m=a}^{b} \Omega_m^\dagger \Omega_m \rho\right)$$

The positive operator $M = \sum_{m=a}^{b} \Omega_m^\dagger \Omega_m$ thus determines the probability that m lies in the subset $[a,b]$ of its range.

Assume the eigenvalues n of the observable N take on all the integer values.

$$\Omega_m = \frac{1}{\mathcal{N}} \sum_n \exp\left[\frac{-k(n-m)^2}{4}\right] |n\rangle\langle n|$$

$\hookrightarrow$ s.t. $\sum_{m=-\infty}^{\infty} \Omega_m^\dagger \Omega_m = \mathbb{1}$

Examine the case where we start with no information about the system. In this case the density matrix is completely mixed, so $\rho \propto \mathbb{1}$.

After making the measurement and obtaining the result m, the state of the system is

$$\rho_f = \frac{\Omega_m \rho \Omega_m^\dagger}{Tr[\Omega_m \rho \Omega_m^\dagger]} = \frac{1}{\mathcal{N}} \sum_n \exp\left[\frac{-k(n-m)^2}{2}\right] |n\rangle\langle n|$$

peaked about the eigenvalue m, but has a width given by $\frac{1}{\sqrt{k}}$.

k larger $\longrightarrow$ less uncertainty.

$$\begin{cases} \text{large } k : & \text{strong measurement} \\ \text{small } k : & \text{weak measurement} \end{cases}$$

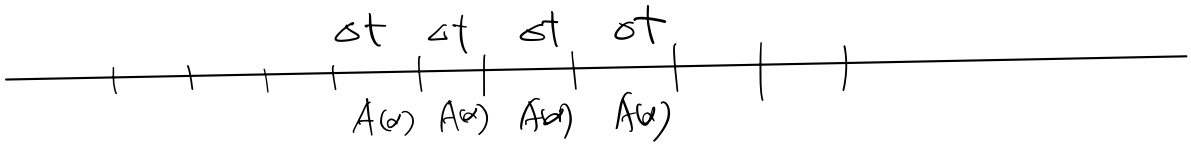
A conti. measurement of an observable

"information is continually extracted from a system"

To construct a measurement like this, we can divide time into a sequence of intervals of length Δt, and consider a weak measurement in each interval.

measurement strength $\propto \Delta t$, $\Delta t \to 0$

X : Hermitian, continuous spectrum $\begin{cases} X|x\rangle = x|x\rangle \\ \langle x|x'\rangle = \delta(x-x') \end{cases}$



$$A(\alpha) = \left(\frac{4k\Delta t}{\pi}\right)^{\frac{1}{4}} \int_{-\infty}^{\infty} \exp\left[-2k\Delta t\,(x-\alpha)^2\right] |x\rangle\langle x|\, dx$$

initial state : $|\psi\rangle = \int \psi(x)\,|x\rangle\, dx$, then $P(\alpha) = \mathrm{Tr}\left[A^\dagger(\alpha) A(\alpha) |\psi\rangle\langle\psi|\right]$

$$\langle\alpha\rangle = \int_{-\infty}^{\infty} \alpha\, P(\alpha)\, d\alpha = \int_{-\infty}^{\infty} \alpha\, \mathrm{Tr}\left[A^\dagger(\alpha) A(\alpha) |\psi\rangle\langle\psi|\right] d\alpha$$

$$= \sqrt{\frac{4k\Delta t}{\pi}} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \alpha\, |\psi(x)|^2 \exp\left[-4k\Delta t(x-\alpha)^2\right] dx\, d\alpha$$

$$= \int_{-\infty}^{\infty} x\, |\psi(x)|^2\, dx = \langle X\rangle$$

$$P(\alpha) = \mathrm{Tr}\left[A^\dagger(\alpha) A(\alpha) |\psi\rangle\langle\psi|\right] = \left(\frac{4k\Delta t}{\pi}\right)^{\frac{1}{2}} \int_{-\infty}^{\infty} |\psi(x)|^2 \exp\left(-4k\Delta t(x-\alpha)^2\right) dx$$

If Δt is sufficient small, then the Gaussian is much broader than $\psi(x)$, which means we can approximate $|\psi(x)|^2$ by a delta function, which must be centred at the expected position $\langle\alpha\rangle = \langle X\rangle$.

Hence :

$$p(\alpha) \approx \left(\frac{4k\Delta t}{\pi}\right)^{\frac{1}{2}} \int_{-\infty}^{\infty} \delta(x - \langle X \rangle) \exp\left[-4k\Delta t (x-\alpha)^2\right] dx$$

$$= \left(\frac{4k\Delta t}{\pi}\right)^{\frac{1}{2}} \exp\left[-4k\Delta t (\alpha - \langle X \rangle)^2\right]$$

$$\alpha_s = \langle X \rangle + \frac{\Delta W}{(8k)^{\frac{1}{2}} \Delta t} \qquad \Delta W \sim N(0, \Delta t)$$